

Chih-Yun (Howard) Pai

213-696-2958 | tmps9930715@gmail.com | linkedin.com/in/chiyun-pai/ | https://github.com/Howard-Pai

SKILLS

- **Programming Languages:** Python, SQL, Java, Bash, C, R, JavaScript, HTML, CSS
- **Data Engineering:** Spark, Hadoop, Kafka, ETL pipelines, Distributed Processing
- **Databases & Cloud:** Snowflake, MySQL, MongoDB, Amazon EC2, Google Cloud, Firebase
- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, HuggingFace, RAG, Graph RAG
- **Tools & Systems:** Git, Claude Code, CUDA, Amazon Lambda, Kafka, Linux, Docker, React

EDUCATION

- **University of Southern California** Aug. 2024 - May 2026
Master of Science in Applied Data Science Los Angeles, United State
- **National Chengchi University** Sep. 2019 - Jun. 2023
Bachelor of Arts in Economics with a Minor in Management of Information System Taipei, Taiwan

WORK EXPERIENCE

- **Information Science Institute** Mar. 2026 - present
Volunteer researcher
 - Worked on graph neural network (GNN) regression and classification tasks by translating knowledge graph data prediction problems into GNN-based learning frameworks using KGEDSL.
 - Developed Knowledge Graph Retrieval-Augmented Generation (KGRAG) pipelines that retrieve relevant knowledge subgraphs and leverage LLMs for prediction generation and performance comparison.
- **National Academy of Taiwan** Nov. 2024 - Jun. 2025
Part-time Research Assistant Remote
 - Conducted Retrieval-Augmented Generation (RAG) experiments to analyze the impact of context reordering, fine-tuning strategies, and golden-to-irrelevant context ratios on small-domain dataset.
 - Evaluated long-form answer quality using multiple LLM evaluation metrics, including BERTScore, GPTScore, and G-Eval, to assess semantic similarity, factual consistency, and response coherence.
 - Collaborated with *Innolux Corporation* to investigate the applicability of Graph RAG for enterprise knowledge retrieval and multi-hop reasoning tasks, improving information traceability.
- **Taiwan Institute of Economic Research** Nov. 2022 - Jun. 2023
Assistant Researcher Intern Taipei, Taiwan
 - Analyzed industrial and policy reports on telecommunication-related topics, including 5G infrastructure development, digital transformation, broadband policy, and emerging communication technologies.
 - Generated research summaries and regulatory analyses for the National Communications Commission (NCC) to support policy evaluation, telecommunications market assessment, and industry monitoring.
 - Performed create, read, update, delete (CRUD) operations on internal databases and maintained data integrity through official API to support reliable data synchronization and accurate record management.

PROJECTS

- **OCR-Based Data Pipeline for Oil well data | PyTesseract, Snowflake, Vite** Spring 2026
 - Built an end-to-end ETL pipeline to extract structured data from scanned PDF documents using OCR.
 - Automated batch ingestion of unstructured files and transformed extracted text into structured tables.
 - Enriched datasets by integrating external web data through automated scraping and API-based queries.
- **Kalman Filter Pair Trading Strategy | Python, JAX, PySpark, Kafka** June 2025 - Present
 - Implemented 4-layer Kalman Filter in JAX (JIT) for real-time beta estimation and spread generation.
 - Built Level 2 tick data pipeline (Databento + Kafka) and applied OPTICS clustering on 500+ equities.
 - Developed adaptive beta system with AR(1)-based regime detection for robust hedge ratios.